Ghost Job Detection: A Labor Market Mispricing Signal

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1. Introduction

Financial Problem and Motivation

Job postings are widely used as leading indicators of corporate health. Analysts and investors treat hiring activity as a proxy for firm-level investment intent and future earnings (Gutiérrez et al., 2020). Yet an estimated 21% of active postings are ghost jobs — legitimate listings maintained by real firms with no near-term intent to hire (Ng, 2024) — and surveys suggest 40% of companies have deliberately done so. The mispricing implication is direct: if markets systematically read ghost postings as growth signals, ghost-heavy firms will be temporarily overvalued relative to their fundamental hiring trajectories, creating a potential source of cross-sectional alpha.

Theoretical Framework

We organise expected return dynamics around two competing hypotheses. **H1 (Fundamental Mispricing)** predicts that ghost-heavy firms earn negative long-run abnormal returns as earnings disappointments are eventually priced in. **H2 (Attention and Behavioural Premium)** predicts a sustained positive premium driven by investor attention to high-posting firms, consistent with sentiment-driven demand and limits to arbitrage that prevent rapid correction. The hypotheses are not mutually exclusive: H2 may dominate at quarterly horizons while H1 operates on a 3–5 year horizon.

Literature Context

Three papers directly motivate our approach. Ng (2024) uses BERT to classify approximately 21% of Glassdoor-reviewed postings as ghost jobs, but relies on sentiment-proxied labels without confirmed hiring outcomes or financial return implications. Amaar et al. (2022) achieve 99.9% accuracy on fraudulent scam-posting datasets — a conceptually distinct target from legitimate firms posting ghost jobs. Singer and Oktay (2025) propose a behavioral feature framework for ghost detection but provide no trained model and draw no financial conclusions. Our contribution is to implement this framework end-to-end using a supervised regression model trained on confirmed hiring outcomes, validate it through Gaussian Mixture Models (GMM), and demonstrate its translation into a walk-forward trading signal.

2. Methodology

Data

We use Revelio Labs workforce intelligence data: 310,821 deduplicated job postings across 2,919 US public firms from 2009–2024. After filtering to firms with at least three postings per quarter, 1,017 firms enter the model across 7,167 training, 884 validation, and 886 test firm-quarter observations.

Features and Model

We construct 14 firm-quarter features (Table 1): ten capture posting behavior — volume, average and median duration, share never removed (`still_active_rate`), salary disclosure rate, average posted salary, Revelio's expected hire estimates, role diversity, and remote rate — and four capture firm fundamentals: headcount, YoY headcount growth, log-headcount, and NAICS sector. The outcome variable is confirmed employee inflows in the following quarter.

#	Feature	Economic rationale
1	`num_postings`	High posting volume relative to hires signals low conversion intent
2–3	`avg/median_duration`	Long-duration postings indicate postings are not being actively filled
4	`still_active_rate`	Postings never removed — the primary ghost behavioral signal
5	`salary_disclosure_rate`	Low disclosure signals reduced commitment to filling the role
6	`avg_salary`	Higher salary postings associate with genuine, competitive hiring
7–8	`total/avg_expected_hires`	Revelio's model-based hire expectations; controls for stated intent
9	`role_diversity`	Low diversity suggests repeated ghost posting of the same role
10	`remote_rate`	Context control for posting strategy differences
11–12	`headcount, log_headcount`	Primary size controls; top two SHAP drivers
13	`headcount_growth_4q`	YoY growth captures genuine organic hiring demand
14	`naics_int`	Sector baseline hiring rates vary across industries

Table 1. Feature matrix. Features 1–10: posting behavioral; 11–14: firm fundamentals.

An XGBoost regressor predicts next-quarter inflows from the 14 features, benchmarked against Ridge regression. Early stopping (best iteration: 185) is applied on the validation set. The ghost intensity score is defined as:

```
GhostScore(i, t) = 1 − Rank_t[PredFillRate(i, t)] / N_t
```

where `PredFillRate(i,t)` = XGBoost[X(i,t)] / Postings(i,t), predicting inflows at t+1 with no lookahead. Cross-sectional percentile ranking absorbs firm-size effects so the score captures deviation from expected hiring. Firms below the 21st percentile threshold (adopted from Ng 2024) are flagged as ghost-heavy: 43 of 204 firms (21.1%) in the test set. A GMM on the same 14 features without labels confirms the structure: BIC selects k=7; ghost-prone clusters 3 and 4 exhibit ghost rates of 44% and 50% respectively; Cluster 0 (most genuine-hirer) shows only 1.6% (see Appendix A for BIC model selection and cluster profile heatmap).

Backtest Design

Six strategies are evaluated in a walk-forward framework with 20 bps transaction cost: Long High Ghost (most ghost-intensive decile), Long Low Ghost (genuine hirers), Ghost Barbell (both extremes), Long/Short (paired legs), Ghost Momentum (changes in intensity), and Threshold (binary flag). Concentration levels of 5%, 10%, and 25% are tested. SPY is the passive benchmark; an equal-weight universe portfolio is a sampling-bias diagnostic.

3. Results

Model Performance

XGBoost achieves test R^2=0.486 (MAE=3.1), substantially outperforming Ridge (R^2=0.279). Prediction accuracy is secondary to ranking quality: the score's purpose is to correctly order firms by ghost intensity, not to forecast precise hire counts. The ghost score distribution is right-skewed with most firms near zero and a thin high-intensity tail (Figure 1). The most ghost-like firms — BFAM, CPRT, HRB, COKE, SRCL — are established consumer-service companies with year-round posting pipelines but near-zero conversion. Genuine hirers at the other extreme — EXPI, F, LUV, DAL, BX — convert postings at rates approaching or exceeding one hire per posting per quarter (see Appendix B for full top-20 and bottom-20 firm-level rankings).

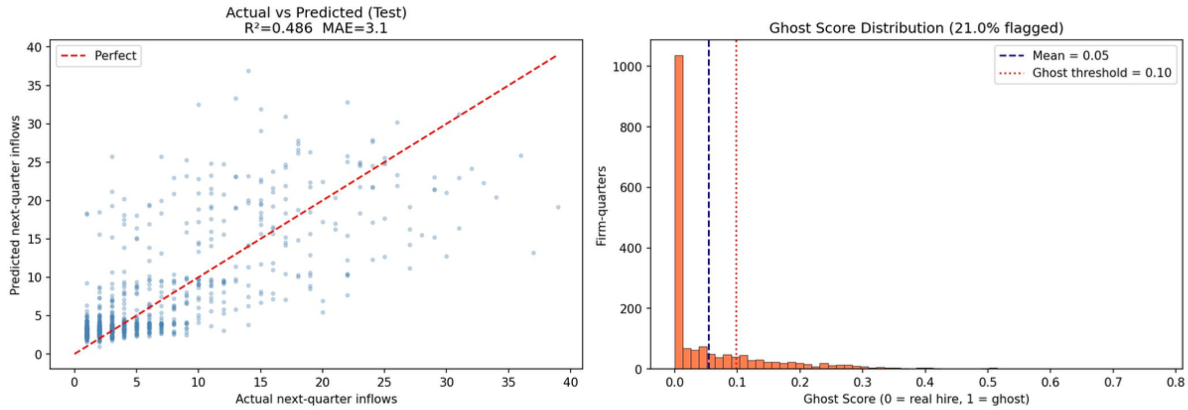

Figure 1. Left: Actual vs. predicted inflows on the test set (R^2=0.486, MAE=3.1). Right: Ghost score distribution across 5,940 firm-quarters; 21.0% exceed the threshold of 0.10.

SHAP Attribution

SHAP beeswarm analysis on 886 test observations (Figure 2) reveals headcount (mean|SHAP|=4.45) and log-headcount (2.15) dominate predictions — the mechanical baseline of larger firms hiring more. The cross-sectional ranking design embeds size effects in the baseline so the score captures deviations, not absolute volume. Among ghost-specific features, high `still_active_rate` values (blue dots left of zero) consistently push predicted inflows downward, confirming that firms that never remove postings are correctly identified as low-conversion filers. The behavioral ghost signal is directionally sound, appropriately scaled relative to firm-size effects.

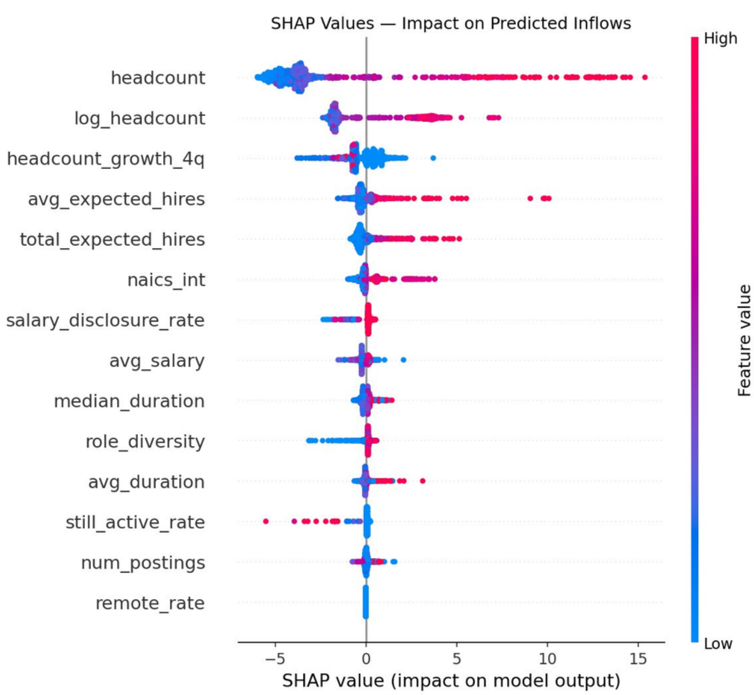

Figure 2. SHAP beeswarm on 886 test observations. Red = high feature value; blue = low. Dots left of zero reduce predicted inflows. still_active_rate (high values, blue, left) is the clearest ghost behavioral signal.

Strategy Performance

Table 2 summarises walk-forward performance at 10% concentration over 2019–2026. Ghost Barbell (+216.9%, Sharpe 0.84) outperforms SPY (+180.1%, Sharpe 0.82) on both return and risk-adjusted dimensions. Long High Ghost (~220%) delivers the highest raw return, consistent with H2: investor attention to high-posting firms sustains a medium-term premium. Results are robust across 5%, 10%, and 25% concentration. The Long/Short strategy (–22.1%, Sharpe –0.32) fails — market-wide drift overwhelms the short signal. The equal-weight universe (+139.4%) underperforms SPY, confirming the alpha is not a sampling artifact.

Strategy	Total Return	CAGR	Sharpe	Max DD	Interpretation
Long High Ghost	~220%	—	—	—	*H2: attention premium + growth narrative extrapolation*
Ghost Barbell	**+216.9%**	+17.1%	**0.84**	–37.4%	Best risk-adjusted; beats SPY on both return and Sharpe
Long Low Ghost	~120%	—	—	—	*Genuine hirers outperform long-run; quality signal*
SPY (benchmark)	**+180.1%**	+15.2%	0.82	–33.7%	Passive benchmark
Universe (EW)	+139.4%	+12.7%	0.64	–39.4%	*Trails SPY; confirms alpha not a sampling artifact*
Long / Short	**–22.1%**	**–3.4%**	**–0.32**	**–43.4%**	*Short leg overwhelmed by bull-market drift*

Table 2. Strategy performance at 10% concentration, 2019–2026. — = chart-derived estimates; full metrics not available in truncated output.

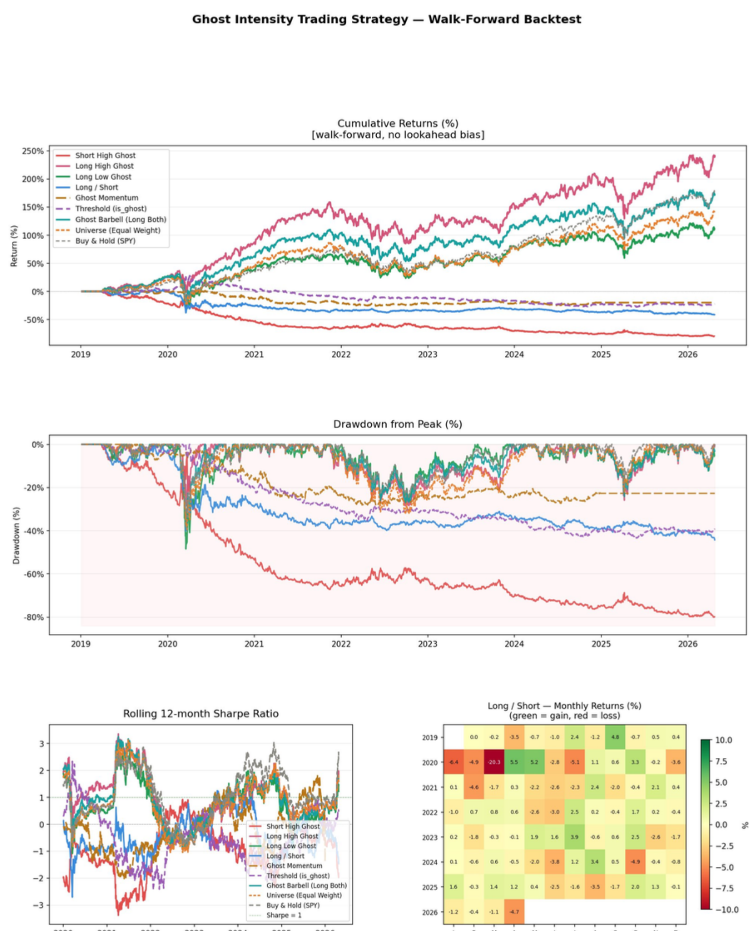

Figure 3. Walk-forward backtest cumulative returns, 2019–2026. Ghost Barbell and Long High Ghost consistently outperform SPY. Short strategies exhibit persistent drawdown.

4. Critical Evaluation

Limitations and Model Stability

The test R^2 of 0.486 is appropriate for a ranking signal; the train-to-test drop (0.947 to 0.486) indicates overfitting that early stopping partially addresses, but walk-forward no-lookahead discipline ensures strategy performance is not in-sample inflated. GMM stability holds across k=4–8.

- **Hire proxy noise.** Confirmed start-date records undercount actual hires; cross-sectional rankings are more reliable than absolute ghost scores.
- **Market regime dependence.** The 2019–2026 period is predominantly a bull market; the H2 attention premium is likely regime-dependent and bear-market performance is untested.
- **Short-leg failure.** Long/Short collapse (–22.1%, Sharpe –0.32, max DD –43.4%) confirms that naked shorting ghost firms requires factor-neutral construction hedging market beta and sector exposures.
- **Platform bias.** Only Revelio-covered firms are observable; referral-heavy recruiters are excluded, potentially inflating ghost rate estimates.

Interpretation, Decision Relevance, and Contribution

The most consequential finding requiring honest interpretation is that Long High Ghost outperforms Long Low Ghost, contradicting H1. We interpret this as H2 operating over the medium-term backtest horizon combined with a horizon mismatch: the fundamental correction under H1 likely materialises over 3–5 years involving earnings misses and hiring stalls, while quarterly rebalancing captures only the persistence phase. Testing H1 requires longer event studies with factor-controlled returns — a clear next step alongside Fama-MacBeth regressions and factor-neutral long-short construction.

Ghost Barbell (+216.9%, Sharpe 0.84 vs. SPY Sharpe 0.82) demonstrates tradeable alpha applicable as a portfolio overlay or as an additional factor alongside value, size, and momentum. The signal also has monitoring value: a sustained increase in ghost intensity may predict earnings disappointments before analyst consensus updates. This work is the first to link ghost-job detection to financial return outcomes through a validated, outcome-anchored signal — prior work relies on sentiment proxies (Ng, 2024), targets fraudulent scam postings (Amaar et al., 2022), or stops at the framework stage (Singer & Oktay, 2025). The GMM independently confirms that ghost behavior is structurally real, demonstrating that corporate behavioral patterns — how firms post, not just what they post — contain pricing-relevant information not yet fully reflected in equity markets.

References

Amaar, A., Asghar, M., Alam, I., & Javied, A. (2022). Detection of fake job posts using TF-IDF and machine learning classifiers. *Zenodo.* https://doi.org/10.5281/zenodo.6810048

Gutiérrez, G., Jones, C., & Philippon, T. (2020). Entry costs and the macroeconomy. *Management Science*, *67*(10), 6320–6356. https://doi.org/10.1287/mnsc.2020.3636

Ng, H. (2024). *Ghost job detection via BERT on Glassdoor reviews* [Working paper].

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Appendix A — GMM Model Selection and Cluster Profiles

Figure A1 shows BIC and AIC information criteria across k=2 to k=8 clusters. Both metrics reach their minimum at k=7 (BIC=−1,508), confirming the optimal number of behavioral clusters without supervision. Figure A2 presents z-scored feature means per cluster, revealing distinct behavioral profiles. Cluster 0 (high avg_salary, salary_disclosure_rate, avg_expected_hires) is the most genuine-hirer group. Clusters 3 and 4 exhibit the clearest ghost profiles: long avg_duration, low salary_disclosure_rate, and high predicted_fill_rate deviations. Cluster 5 shows exceptionally high headcount and headcount_growth — the large-volume genuine hirer. The GMM thus independently replicates the ghost-vs-genuine distinction without any supervised label.

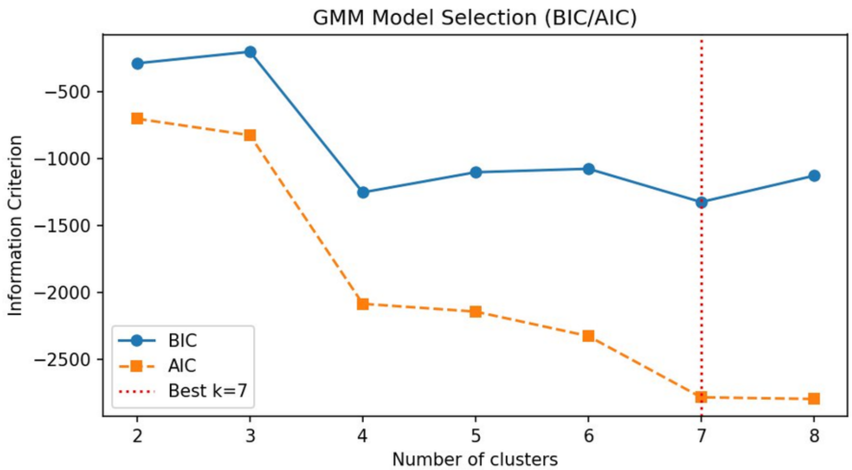

Figure A1. GMM model selection (BIC and AIC) across k=2–8. Both criteria minimise at k=7 (BIC=−1,508), confirming the optimal cluster count.

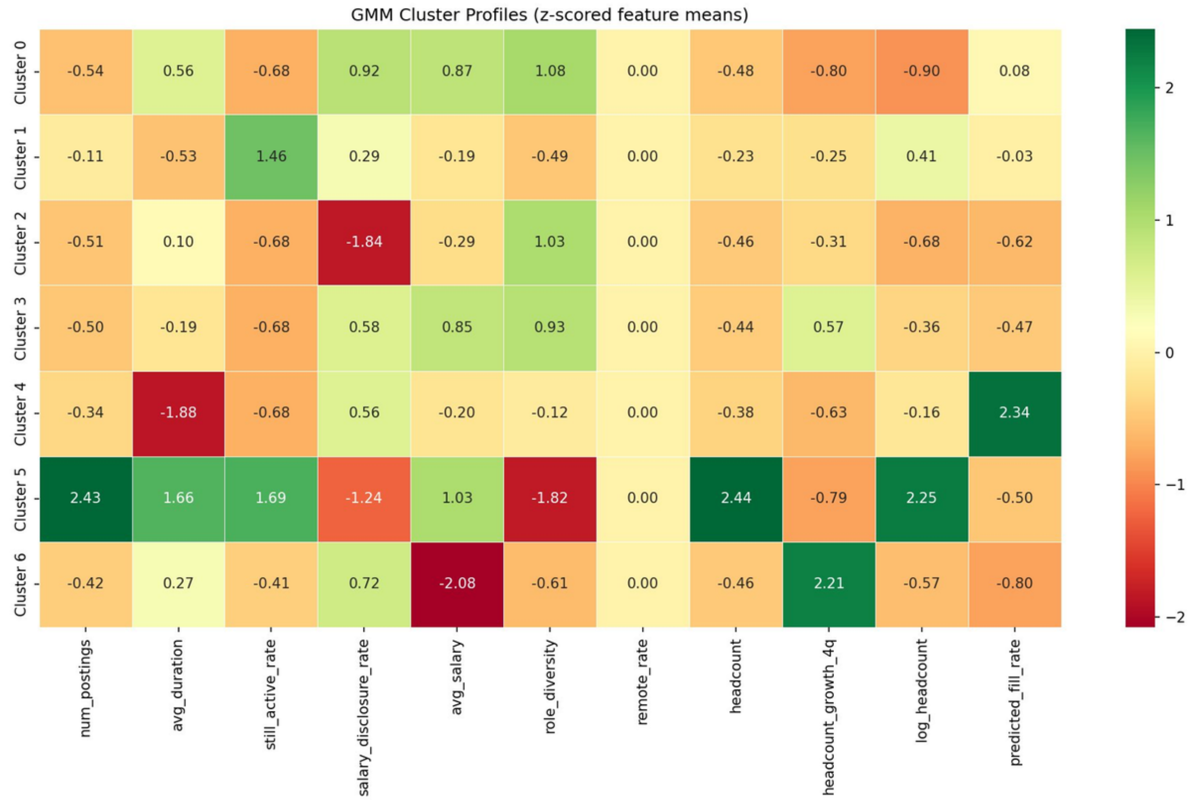

Figure A2. GMM cluster profiles: z-scored feature means per cluster. Green = high relative to cross-sectional mean; red = low. Clusters 3 and 4 show the clearest ghost profiles (long duration, low salary disclosure). Cluster 0 is the most genuine-hirer group.

Appendix B — Ghost Intensity Rankings: Full Firm-Level Charts

Figures A3 and A4 show actual (blue) versus predicted (orange) fill rates for the 20 most ghost-like and 20 most genuine-hirer firms in the test set, ranked by predicted fill rate. Ghost-like firms (Figure A3) are predominantly established consumer-services, healthcare services, and technology companies — AZO, PRFT, THC, DELL, CFG — posting aggressively but converting almost no postings into confirmed hires. The divergence between blue (actual) and orange (predicted) bars for several firms (e.g., CVNA, CNC) reflects the model's ability to detect structural low-fill behaviour even when short-term actual fill rates temporarily spike. Genuine hirers (Figure A4) — led by EXPI, F, LUV, DAL — show predicted fill rates well above 1.0, confirmed by high actual rates, validating that the model correctly captures high-conversion firms at the other extreme.

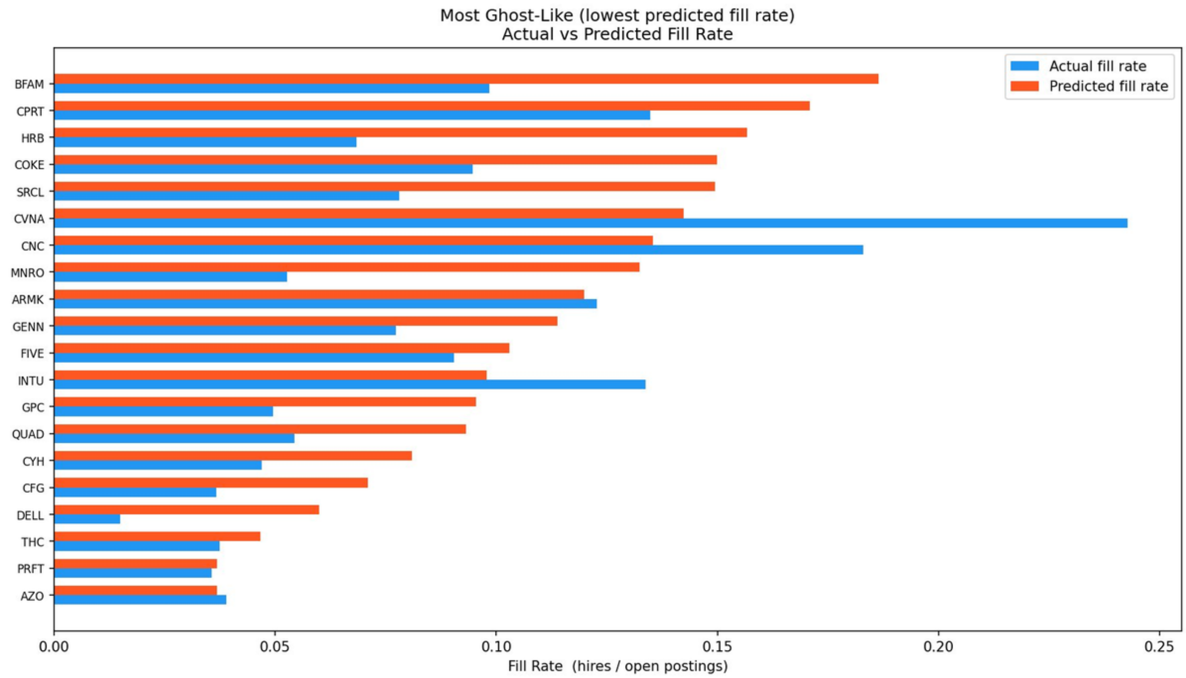

Figure A3. Most ghost-like firms (lowest predicted fill rate). Blue = actual fill rate; orange = predicted fill rate. Firms are ranked from highest ghost intensity (top) to lower.

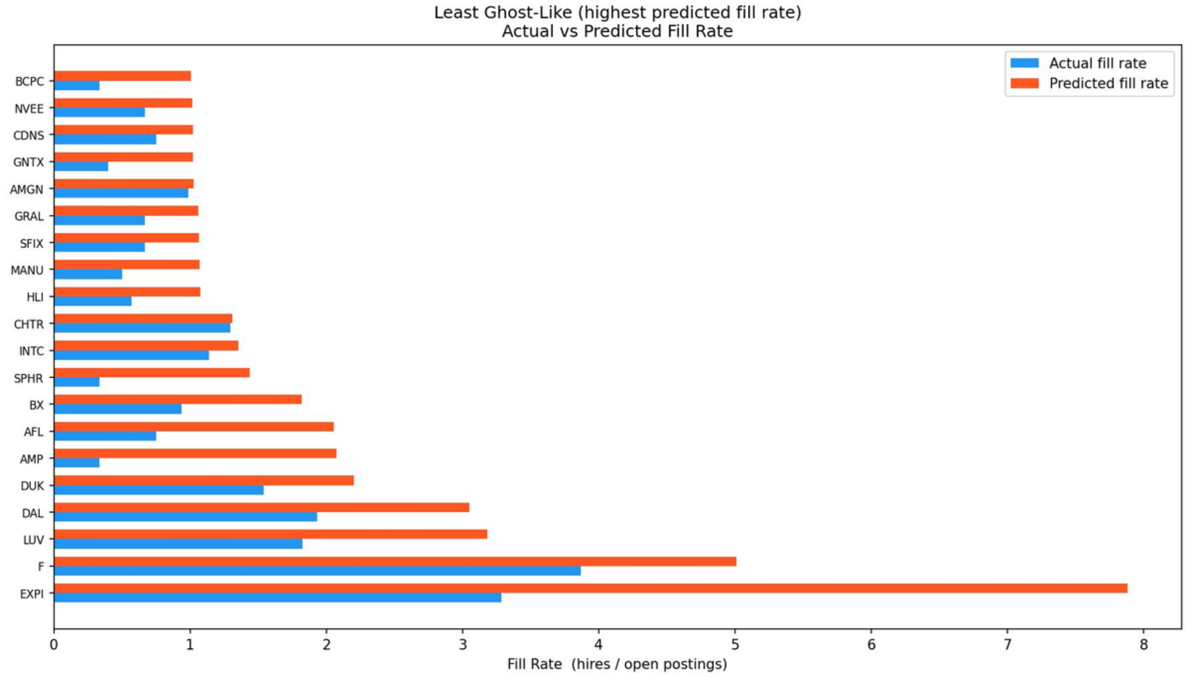

Figure A4. Most genuine-hirer firms (highest predicted fill rate). Blue = actual fill rate; orange = predicted fill rate. EXPI and F show predicted fill rates near 5–8× the ghost firms.